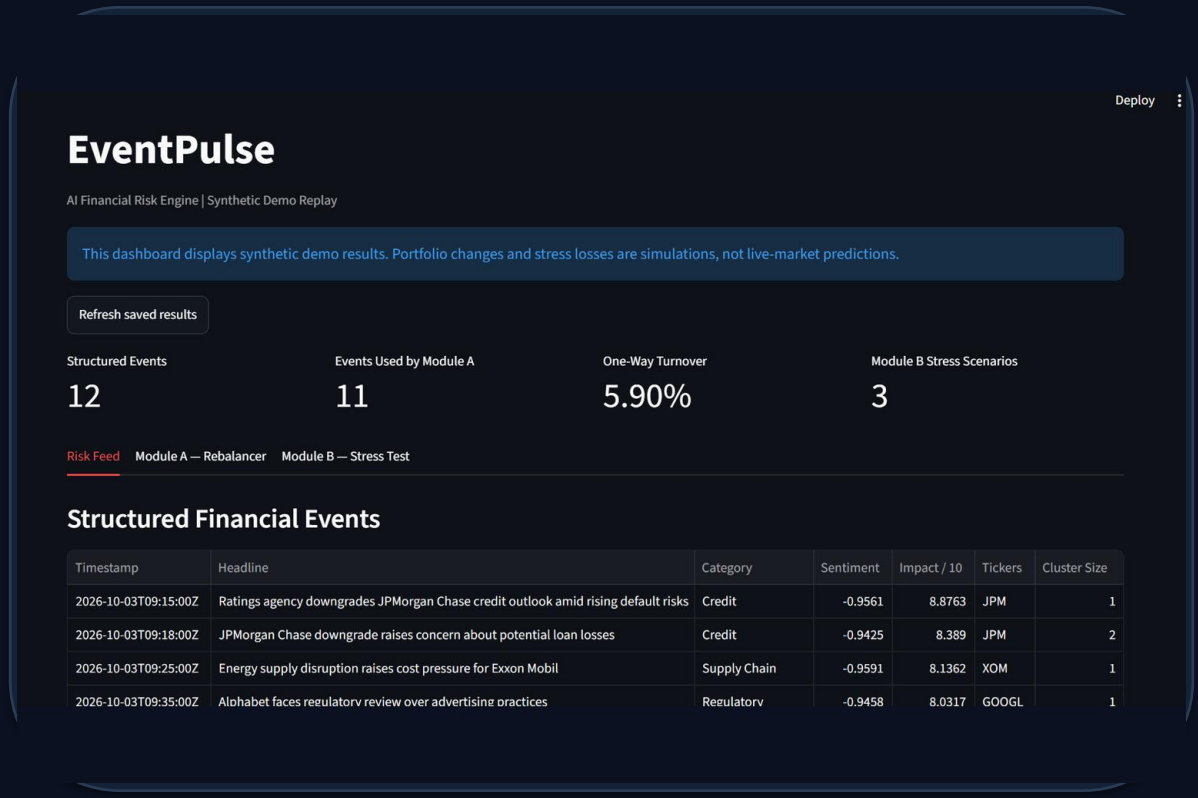
S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

EventPulse

From headlines to events to impact to action.

An AI financial-risk intelligence prototype that turns public news and filings into structured signals for portfolio rebalancing and scenario-based stress testing.

PRETRAINED NLP | EVENT ROUTING | RISK SIMULATIONS



Vansh Goyal
IIIT Delhi

89 tests passed | 2 downstream modules

LIVE INGESTION
GDELT + SEC EDGAR

SYNTHETIC REPLAY
Reproducible demo

The gap is not sentiment - it is turning text into decisions.

Unstructured information becomes useful when events, entities, confidence and downstream rules share one contract.

01 / INPUT OVERLOAD

News is unstructured

Different feeds use different schemas, timestamps and levels of detail.

02 / DUPLICATE COVERAGE

Same event, many reports

Counting repeated headlines as independent evidence can exaggerate a story.

03 / ACTION GAP

Signals stop at a score

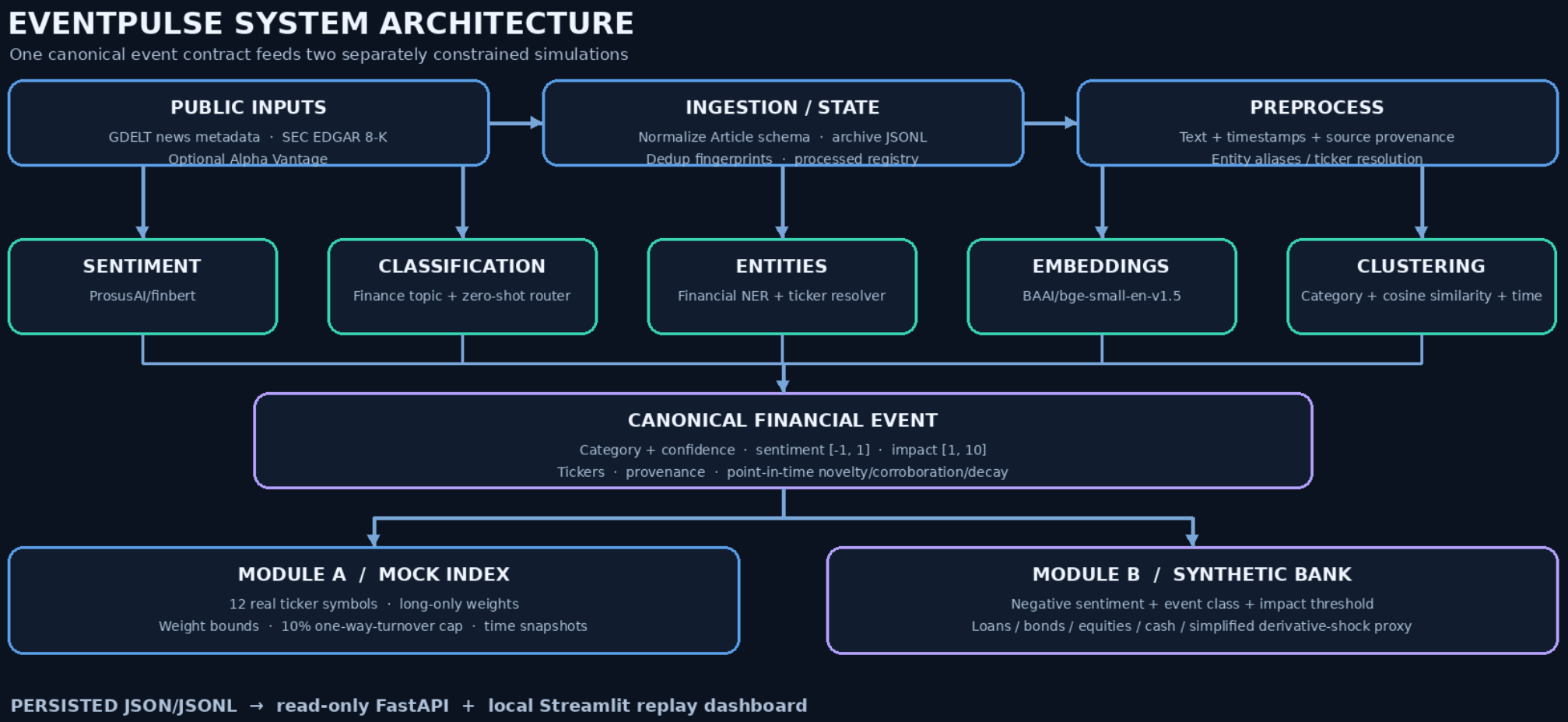
A sentiment label alone does not identify a holding or trigger a stress scenario.

EVENTPULSE APPROACH

01	INGEST	GDELT + SEC EDGAR Optional Alpha Vantage
02	UNDERSTAND	FinBERT + NER + event router Embeddings + semantic clusters
03	STRUCTURE	Canonical FinancialEvent Provenance + confidence + impact
04	SIMULATE	Module A: weights Module B: stress scenarios

One event contract powers both risk modules.

Sources -> NLP -> structured signals -> two constrained simulations. Historical impact training is optional, not active.



Specialized models. Explicit routing. Interpretable rules.

Each component has one role; downstream decisions remain inspectable.

COMPONENT	WHAT IT DOES	WHY IT FITS
FinBERT ProsusAI/finbert	Financial sentiment Signed score in [-1, 1]	Financial-domain language rather than generic sentiment.
Financial NER whataboutyou-ai/financial_bert	Entity mentions Organization / person / location	Helps connect company names to ticker-resolution logic.
Finance-topic model leonas5555/finnews-topic-single-classify	Pretrained financial topic Mapped to EventPulse labels	Provides a domain-specific first classification signal.
Zero-shot + router ModernBERT-base-zeroshot-v2.0 + EventClassifierRouter	Compares candidate event labels Router considers agreement and score margin	Adds task-specific label descriptions without training a new classifier.
BGE embeddings BAAI/bge-small-en-v1.5	Normalized text vectors Cosine similarity + time/ticker rules	Groups semantically related reports; embeddings alone do not prove same event.

IMPACT SCORE | ACTIVE PATH **Severity + sentiment strength + breadth + confidence + novelty -> heuristic score 1-10**

No accepted historical impact-model artifact is included; the 1-10 score is not a calibrated probability or return forecast.

A working dashboard exposes the pipeline end to end.

The Streamlit interface reads the saved synthetic replay and makes both downstream modules inspectable.

Deploy

EventPulse

AI Financial Risk Engine | Synthetic Demo Replay

This dashboard displays synthetic demo results. Portfolio changes and stress losses are simulations, not live-market predictions.

Refresh saved results

Structured Events	Events Used by Module A	One-Way Turnover	Module B Stress Scenarios
12	11	5.90%	3

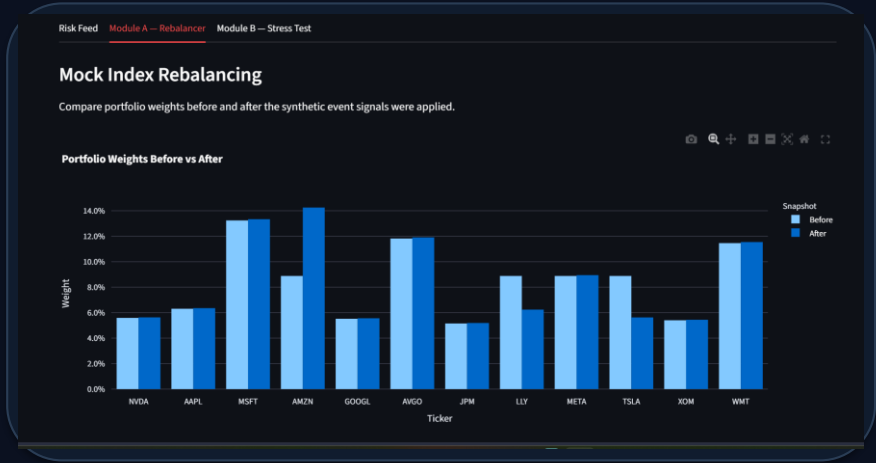
Risk Feed | Module A — Rebalancer | Module B — Stress Test

Structured Financial Events

Timestamp	Headline	Category	Sentiment	Impact / 10	Tickers	Cluster Size
2026-10-03T09:15:00Z	Ratings agency downgrades JPMorgan Chase credit outlook amid rising default risks	Credit	-0.9561	8.8763	JPM	1
2026-10-03T09:18:00Z	JPMorgan Chase downgrade raises concern about potential loan losses	Credit	-0.9425	8.389	JPM	2
2026-10-03T09:25:00Z	Energy supply disruption raises cost pressure for Exxon Mobil	Supply Chain	-0.9591	8.1362	XOM	1
2026-10-03T09:35:00Z	Alphabet faces regulatory review over advertising practices	Regulatory	-0.9458	8.0317	GOOGL	1

RISK FEED / 12 structured events

Headline, category, sentiment, impact, ticker and cluster context in one view.



MODULE A / Before–after portfolio weights



MODULE B / USD 1B portfolio; $76.944M simulated loss

Captures show the current UI; headlines, weights and shocks are synthetic.

From event triage to constrained risk actions.

One event contract connects analyst triage, portfolio review and bank scenario analysis; all replay metrics are synthetic.

12	12	11	5.90%	3
synthetic articles	structured events	replay contributions	one-way turnover	stress results

MODULE A / MOCK-INDEX REBALANCER

Fresh synthetic replay | 12-stock real-ticker universe

LATEST ONE-WAY TURNOVER

5.90% of the configured 10.00% cap

59% of turnover allowance used

12 mock-index holdings
Real ticker symbols
Equal-weight initial state
Not official index weights

11 cluster contributions
processed across the replay
same-cluster reports collapsed

3 rebalance snapshots
Timeline chart now included
Synthetic replay; no live trades

Event strength uses signed sentiment × impact × confidence × novelty × event-specific decay.

Weight changes are constrained; the demo is not a return-maximization backtest.

MODULE A | 12-stock index; 3 snapshots; 5.90% turnover vs 10% cap

MODULE B / SYNTHETIC BANK STRESS TEST

USD-denominated synthetic portfolio | starting value USD 1.000B per scenario

INDEPENDENT SCENARIO RUNS

Credit deterioration | Credit 8.88 — $76.944M
Portfolio after: $923.056M

Supply disruption | Supply Chain 8.14 — $73.880M
Portfolio after: $926.120M

Credit deterioration | Regulatory 8.03 — $76.944M
Portfolio after: $923.056M

The repeated Credit scenario is triggered by two distinct clusters; results are not additive.

MODULE B | Three independent stress results; position-level attribution

89 tests passed (1 dependency deprecation warning). Live smoke: SEC 10 records; GDELT HTTP 429.

Build toward decision-grade risk intelligence.

Future work targets stronger evidence, financial realism and resilient event processing.

CURRENT LIMITATIONS

- Synthetic 12-headline replay; no real-news benchmark or naive-baseline gains measured.
- Impact (1–10) is a heuristic; no accepted trained or calibrated impact model is included.
- The index and bank portfolio are simulated. Derivative exposure uses a return-shock proxy, not pricing or Greeks.
- Public feeds can throttle; GDELT returned HTTP 429 in the last recorded live smoke test.

ENGINEERING ROADMAP

01 **Outcome-linked impact learning**

Build point-in-time event–issuer–market outcome labels; learn impact scores and calibrate confidence without look-ahead leakage.

02 **Financially realistic portfolio controls**

Add transaction costs, liquidity and concentration constraints; model correlated shocks and derivative Greeks/collateral.

03 **Resilient, explainable event intelligence**

Improve issuer/ticker resolution and cross-source corroboration; add source backoff/cache, analyst review and traceable decisions.